INTERNSHIP TASK REPORT

Custom ESP32-S3 SMD PCB Design and Development

Design and Development of a Custom ESP32-S3 Based Surface Mount Device Printed Circuit Board using eSim

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Prepared as part of an Internship / Technical Task involving

eSim Software Ecosystem – IIT Bombay

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Abstract

This report presents the design and development of a custom Surface Mount Device (SMD) Printed Circuit Board (PCB) based on the ESP32-S3-WROOM module. The work was carried out as part of an internship/technical task involving the **eSim software ecosystem developed by IIT Bombay**.

The primary objective of the work was to understand and implement the complete electronic hardware and PCB development workflow, beginning with circuit understanding and schematic development and progressing through component selection, footprint assignment, PCB layout, routing, design verification, and manufacturing preparation.

The PCB integrates a USB Type-C interface, 5 V to 3.3 V power regulation using an AMS1117-3.3 regulator, an ESP32-S3-WROOM module, reset and boot control switches, GPIO expansion, status and test LEDs, decoupling capacitors, and protection components.

Particular attention was given to practical PCB design considerations such as SMD component selection, component availability, power integrity, capacitor placement, trace width, routing length, via usage, grounding, and manufacturability.

The work provided practical exposure to electronic design automation tools and the PCB development workflow. The final design was verified using Electrical Rules Check (ERC), Design Rules Check (DRC), and PCB visualization tools.

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Chapter 1

Project Overview

1.1 Introduction

This work was carried out as part of an internship/technical task involving electronic circuit and PCB design using the **eSim software ecosystem developed by IIT Bombay**.

The objective of the task was to gain practical exposure to the complete workflow involved in electronic circuit design and PCB development.

For the practical implementation, a custom PCB was designed around the ESP32-S3-WROOM module rather than relying on a commercially available development board.

The board was designed to provide the essential hardware infrastructure required to power, program, reset, test, and interface with the ESP32-S3.

The overall work involved understanding and implementing the following stages:

- Circuit understanding and analysis
- Schematic development
- Component selection
- SMD footprint assignment
- PCB component placement
- PCB routing
- Design verification
- Manufacturing preparation

The task provided practical exposure to the workflow followed in electronic hardware development and PCB design.

1.2 Objective

The primary objective of this internship/technical task was to gain practical experience in electronic circuit and PCB development using the eSim IIT Bombay software ecosystem and associated PCB design tools.

1.3 Intended Application

The PCB was developed primarily as a learning and experimentation platform for electronic circuit and SMD PCB design.

The board provides an ESP32-S3 controller with accessible GPIOs and basic testing interfaces, allowing it to be used for future embedded systems and robotics-related experiments.

Chapter 2

Circuit Description

2.1 Overall Circuit Architecture

The PCB receives power and communication through a USB Type-C connector. The USB interface provides a 5 V supply to the PCB. The 5 V supply is then regulated to approximately 3.3 V using an AMS1117-3.3 linear regulator. The regulated 3.3 V supply powers the ESP32-S3 and associated circuitry.

The overall circuit architecture can be represented as:

USB Type-C → 5 V Supply → AMS1117-3.3 → 3.3 V Rail → ESP32-S3 and Peripherals

The ESP32-S3 provides accessible GPIO connections for future external peripherals.

2.2 USB Type-C Interface

A USB Type-C connector is used as the external interface.

The connector provides:

- 5 V VBUS power
- USB D+
- USB D-
- Ground
- Configuration-channel connections

The CC pins use 5.1 kΩ resistors to indicate the appropriate USB Type-C sink configuration.

The USB data lines are connected to the ESP32-S3 USB interface.

Therefore, the USB interface serves both as a power input and as a communication and programming interface.

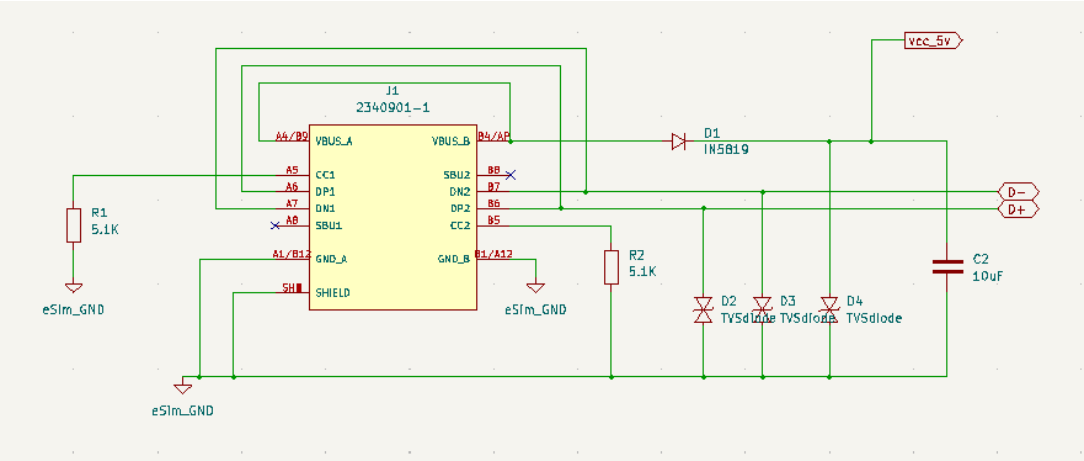


Figure 2.1: USB Type-C Interface Schematic

2.3 USB Protection

TVS protection components were incorporated around the USB interface to provide protection against transient events on the external USB connections.

An additional diode is used in the power path as part of the input power arrangement and protection from reverse current.

The protection circuitry was included because the USB connector is externally accessible and therefore more exposed to electrical disturbances than internal PCB nodes.

2.4 5 V to 3.3 V Power Supply

The USB VBUS provides approximately 5 V to the PCB.

An AMS1117-3.3 linear regulator is used to generate the 3.3 V supply required by the ESP32-S3.

The regulator circuit includes:

- 5 V input
- AMS1117-3.3 voltage regulator
- Input bulk capacitor
- Input high-frequency capacitor
- Output bulk capacitor

- Output high-frequency capacitor
- 3.3 V indicator LED

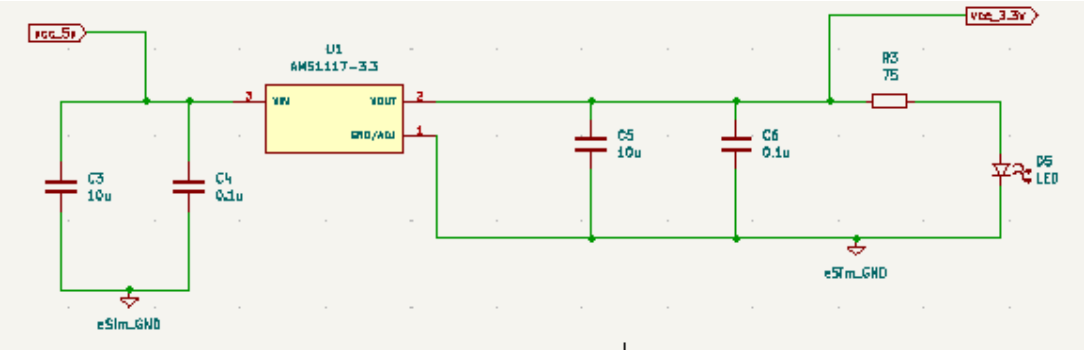


Figure 2.2: 5 V to 3.3 V Regulator Circuit

Chapter 3

Capacitor Selection and Decoupling

3.1 Need for Decoupling Capacitors

Digital integrated circuits such as the ESP32-S3 can experience rapid changes in current demand during switching.

The resistance and inductance of PCB traces prevent the external power supply from responding instantaneously to these changes.

A capacitor placed close to the integrated circuit provides a local charge reservoir and helps reduce voltage fluctuations on the supply rail.

Therefore, capacitors were placed close to the ESP32-S3 power connections.

3.2 Why Multiple Capacitor Values Were Considered

Multiple capacitor values were considered for power filtering and decoupling.

A larger capacitor such as 10 μF provides greater charge storage and is useful for lower-frequency and transient power variations.

Smaller ceramic capacitors such as 100 nF are effective for reducing higher-frequency noise.

Using appropriate capacitor values helps maintain a stable power supply for the ESP32-S3 and associated circuitry.

3.3 Capacitor Placement

The capacitors were deliberately placed close to the ESP32-S3 and regulator power connections.

The objective was to minimize the physical loop area:

Power Pin → Capacitor → Ground

Reducing this loop area helps minimize the effect of parasitic inductance and improves power integrity.

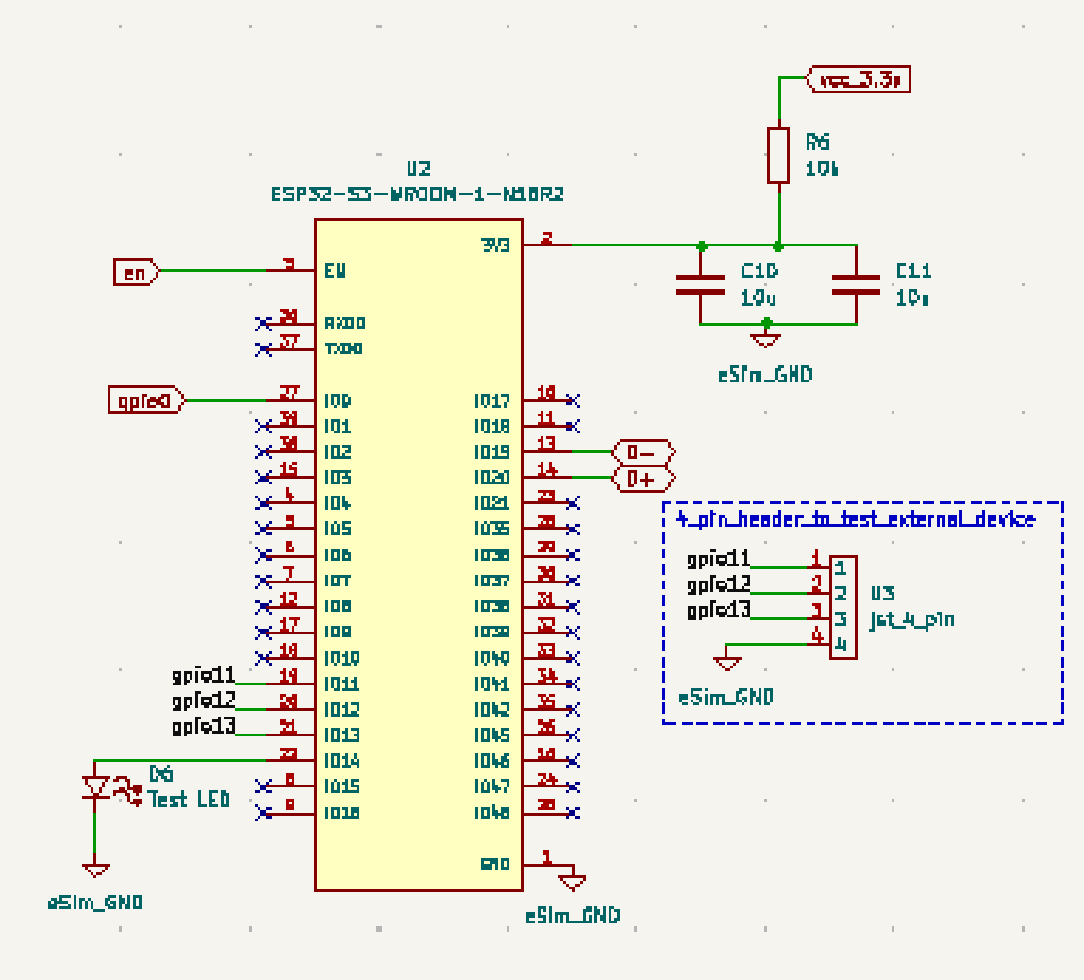


Figure 3.1: ESP32-S3 Decoupling Capacitors

Chapter 4

Reset and Boot Circuit

4.1 Reset Circuit

A dedicated reset switch is provided for manually resetting the ESP32-S3.

The reset circuit uses the EN/CHIP_PU line together with a pull-up resistor and capacitor.

The CHIP_PU pin must not be left floating, and an RC network is used to provide appropriate power-up and reset timing.

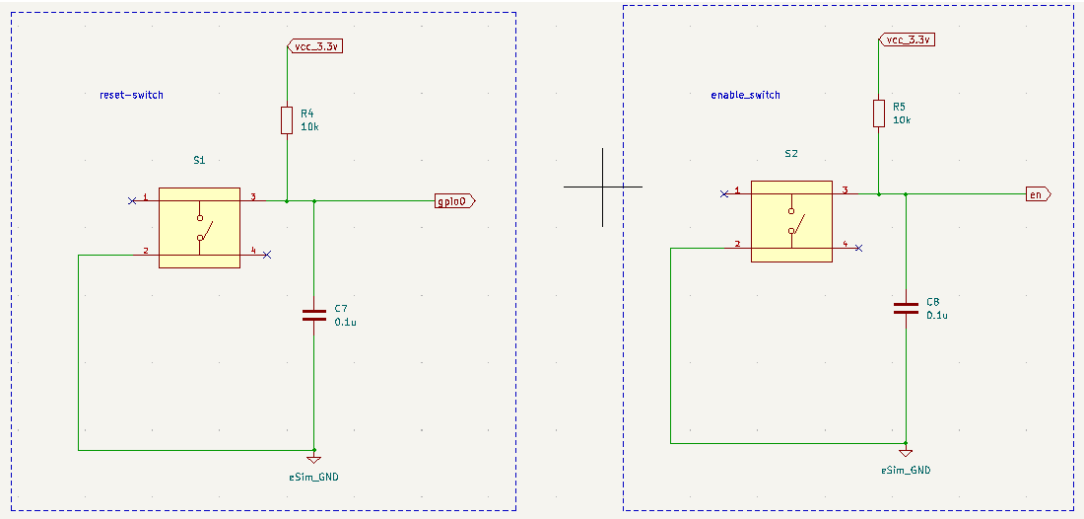


Figure 4.1: ESP32-S3 Reset and Boot Circuit

4.2 Boot Circuit

A dedicated boot switch is provided to control GPIO0.

GPIO0 is an ESP32-S3 strapping pin used during startup to determine the boot mode.

A 10 kΩ pull-up resistor is used to provide a defined logic-high state.

The pull-up resistor provides:

- A defined default logic state
- Stable boot configuration
- Controlled current when the node is pulled low

Chapter 5

Test and Indicator Circuitry

5.1 3.3 V Power Indicator

A dedicated LED is connected to the regulated 3.3 V rail.

Its purpose is to provide a visual indication that the AMS1117 regulator is producing the expected supply voltage.

This provides an immediate hardware-level check before programming the ESP32-S3.

5.2 ESP32 Test LED

A second LED is connected to an ESP32 GPIO.

This LED can be controlled through firmware and can therefore be used to verify:

- The ESP32 is powered.
- Firmware has been successfully uploaded.
- The selected GPIO is functioning.
- The microcontroller is executing code.

This provides a simple first-level functional test of the completed PCB.

Chapter 6

Schematic Design

6.1 Schematic Development

The schematic was developed as part of the electronic design workflow followed during the internship/technical task involving the **eSim IIT Bombay software ecosystem** and associated PCB design tools.

The circuit was organized into logical functional sections to simplify understanding and debugging.

The major schematic blocks are:

- USB Type-C interface
- USB protection
- Power regulation
- ESP32-S3 module
- Reset circuit
- Boot circuit
- GPIO expansion
- LED indicators

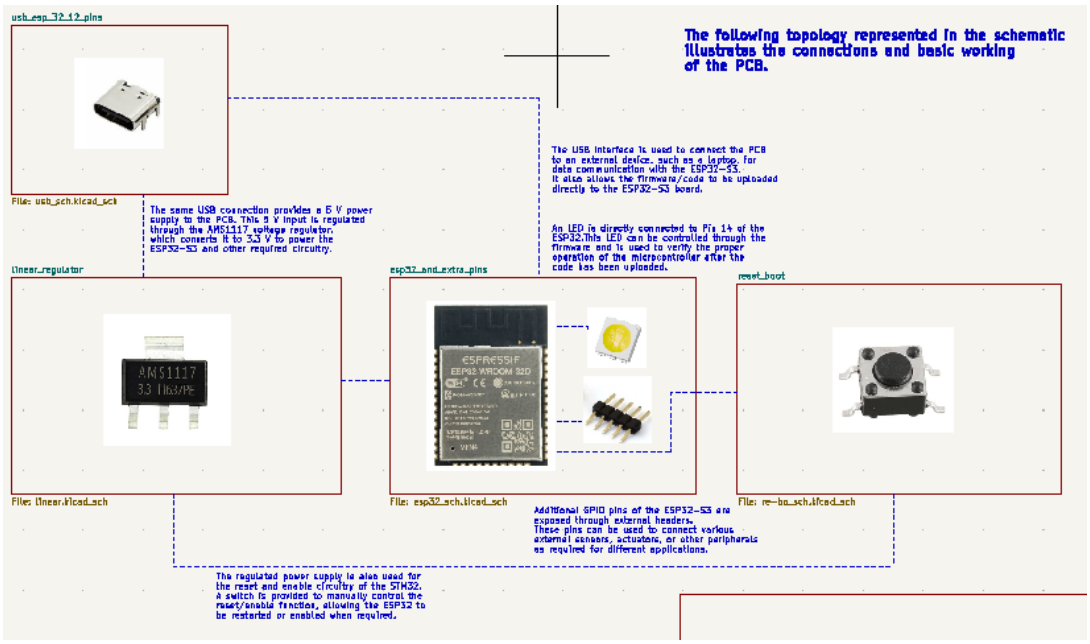


Figure 6.1: Complete System-Level Schematic

Chapter 7

Footprint Assignment

7.1 Footprint Selection Methodology

After completing the schematic, PCB footprints were assigned to the components.

The ESP32-S3 module, switches, and USB Type-C connector required footprints with specific mechanical requirements.

The remaining passive components were selected according to practical SMD package availability.

7.2 Component Market Survey

A physical market survey was conducted at Lamington Road to determine the availability of commonly used SMD components.

The survey helped identify practical resistor and capacitor package sizes that could be sourced locally.

The available components were then considered during footprint selection.

For the passive components, the 0603 package was selected.

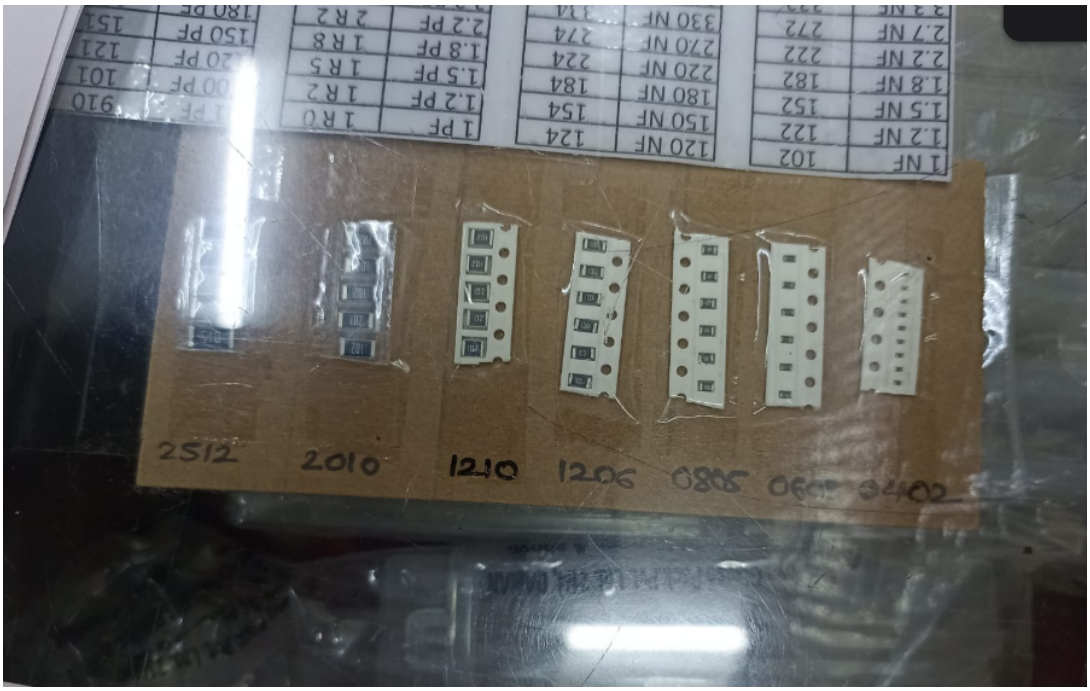


Figure 7.1: Component Market Survey

Chapter 8

PCB Layout

8.1 Board Structure

The PCB was designed as a two-layer board consisting primarily of:

- Top copper layer
- Bottom copper layer
- Solder mask
- Silkscreen
- Board outline

8.2 Component Placement

Component placement was performed before routing.

The major placement considerations were:

- Keep the ESP32-S3 in a mechanically appropriate location.
- Keep decoupling capacitors close to the controller.
- Keep the regulator and power components together.
- Keep the USB connector accessible at the board edge.
- Keep reset and boot switches accessible.
- Minimize unnecessary routing length.
- Reduce the number of unnecessary vias.

- Maintain sufficient spacing between components.
- Keep external connectors accessible.

The ESP32-S3 module was positioned appropriately on the PCB while the power supply, switches, LEDs, and other supporting components were placed to achieve an organized and practical layout.

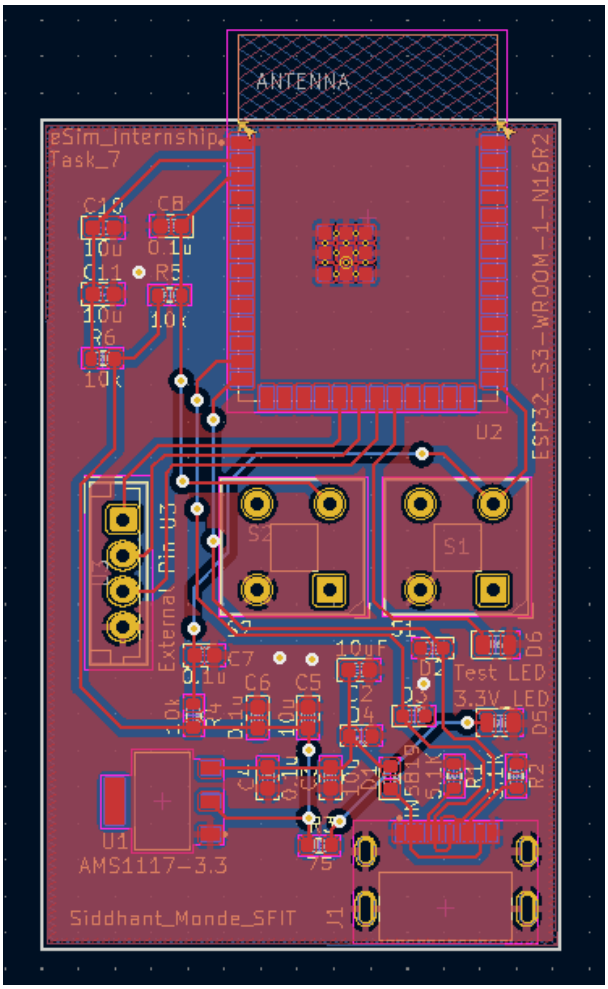


Figure 8.1: PCB Front Copper Layer

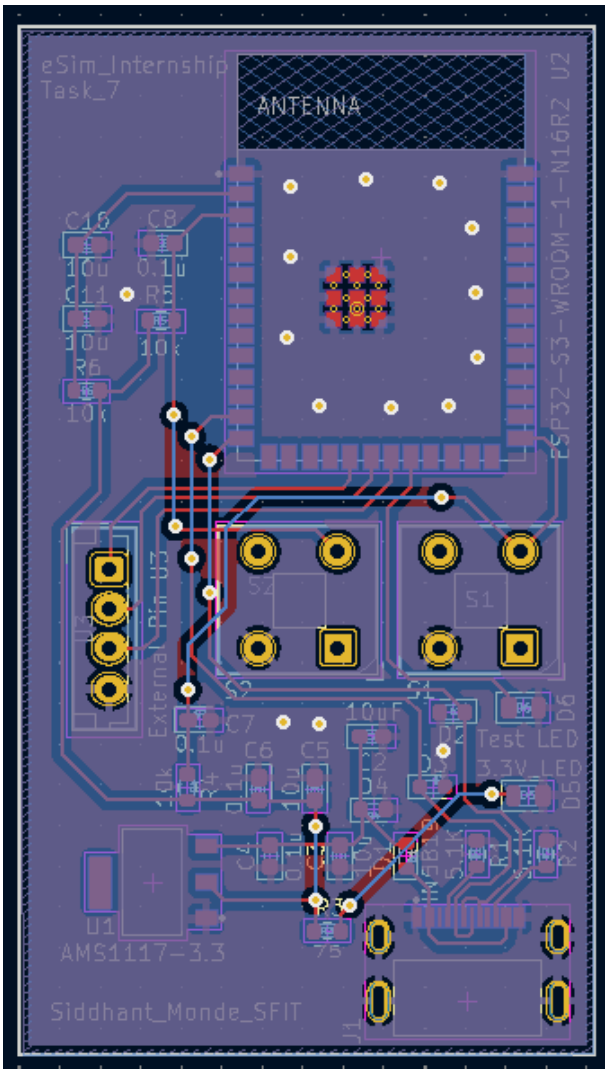


Figure 8.2: PCB Back Copper Layer

Chapter 9

Routing Methodology

9.1 Routing Strategy

Routing was performed with the objective of obtaining short, direct, and manufacturable connections.

The primary routing objectives were:

- Minimum practical trace length
- Minimum unnecessary vias
- Short power paths
- Clean ground return paths
- Adequate clearance
- Easy manufacturability

Critical power connections were routed using wider traces compared with ordinary low-current signal connections.

9.2 Trace Width Selection

A PCB trace-width calculation method based on IPC design methodology was used to estimate an appropriate trace width.

For the general PCB routing:

$$W = 0.25 \text{ mm} \tag{9.1}$$

was selected.

The selected value provides additional manufacturing margin compared with designing directly at the minimum fabrication capability.

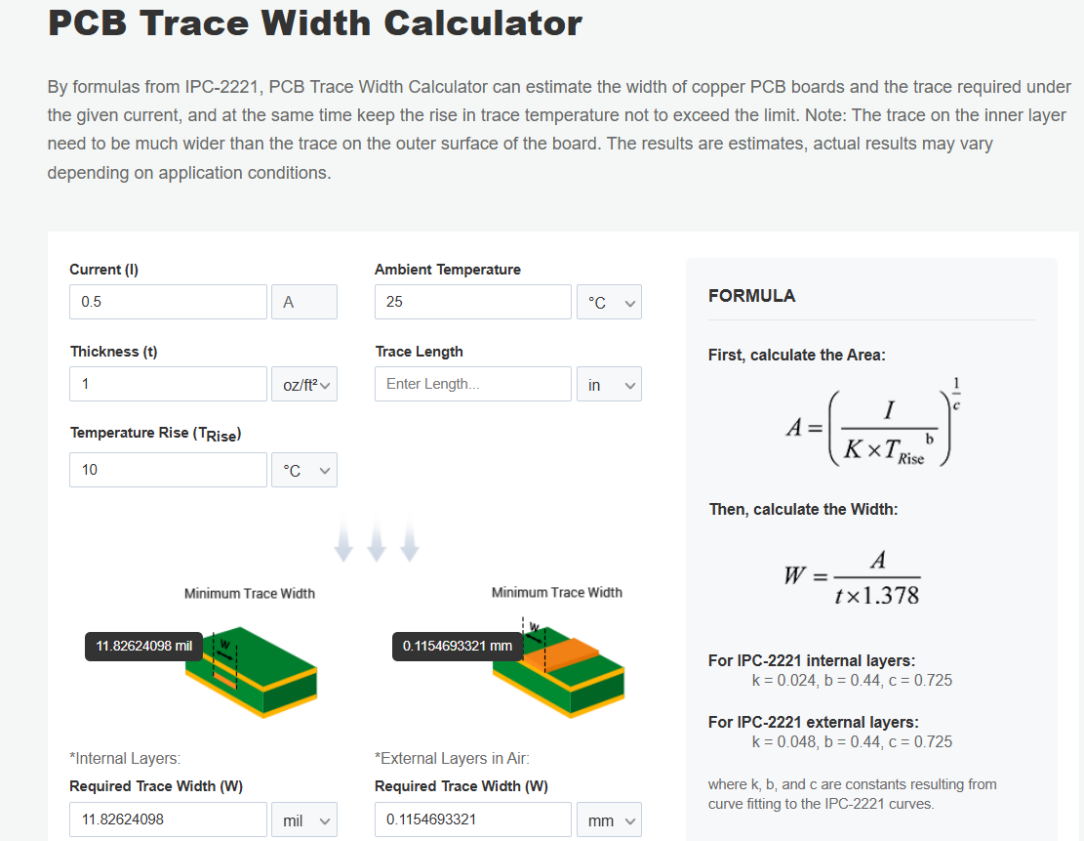


Figure 9.1: Trace Width Calculation

9.3 Via Selection

Vias were used only where required to transition between the top and bottom copper layers.

The number of vias was intentionally minimized because unnecessary vias increase routing complexity and consume useful PCB area.

Via dimensions were selected according to the manufacturing capabilities of the intended PCB fabrication process.

9.4 Ground Polygon

Ground copper polygons were poured on the PCB layers.

The ground planes provide a larger and lower-impedance return path and reduce the amount of dedicated routing required for ground connections.

Maintaining a reasonably continuous ground region is beneficial for power integrity and overall PCB performance.

Chapter 10

Silkscreen Design

The PCB silkscreen was used not only for identification but also as an assembly and debugging aid.

Component references and important component values were included where space permitted.

The advantages include:

- Easier component identification
- Easier assembly
- Easier debugging
- Easier replacement of components
- Verification of component placement
- Better documentation of the board

Resistor and capacitor values were included on the PCB silkscreen where possible.

This was especially useful because the PCB was intended as a learning and manually assembled board.

Chapter 11

Design Rule Check and Verification

11.1 ERC Verification

Electrical Rules Check (ERC) was used to verify the schematic connections.

The verification process was used to identify possible electrical connection problems and warnings associated with the schematic.

Power-related warnings can occur when the software does not recognize a power net as being driven by a valid power-output source.

For a final engineering release, such warnings should preferably be explicitly resolved using appropriate power symbols, power flags, or power-output definitions.

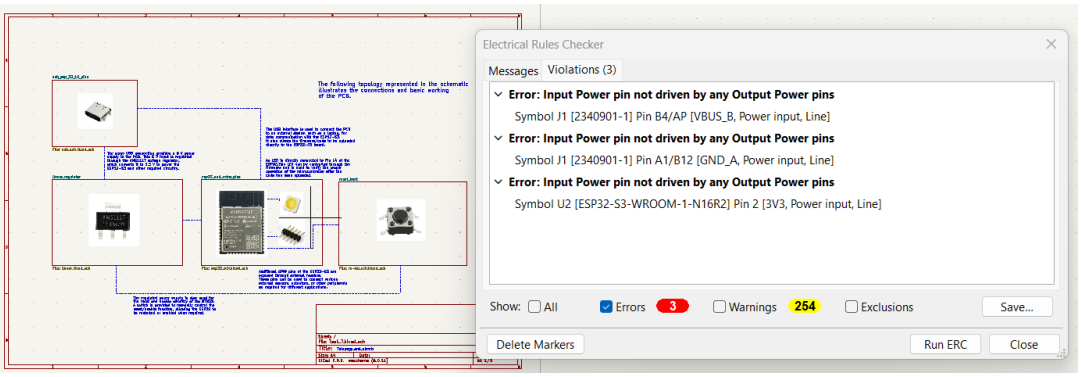


Figure 11.1: ERC Results

11.2 DRC Verification

After PCB routing, the Design Rules Check (DRC) was performed.

The DRC was used to verify:

- Track clearance
- Track width

- Pad clearance
- Via dimensions
- Board-edge clearance
- Unconnected nets
- Copper-zone clearances
- Silkscreen-related constraints

The final PCB design was checked and investigated for design-rule violations before preparing the board for manufacturing.

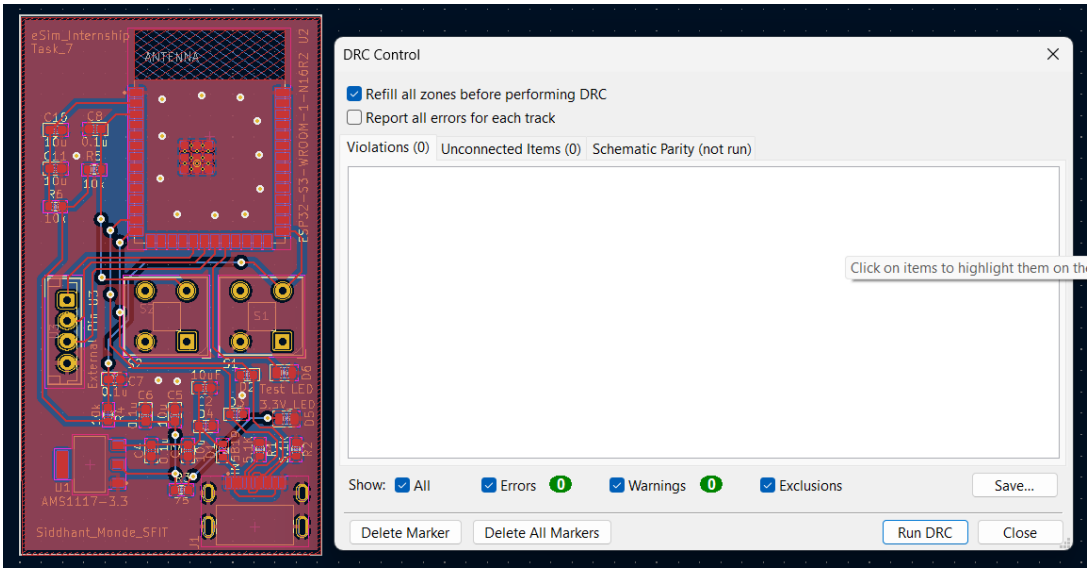


Figure 11.2: Final DRC Verification

Chapter 12

Manufacturing Preview

The generated PCB manufacturing files were uploaded to the manufacturer's online interface to verify the board preview before ordering. And to get a idea on the PCB manufacturing prices.

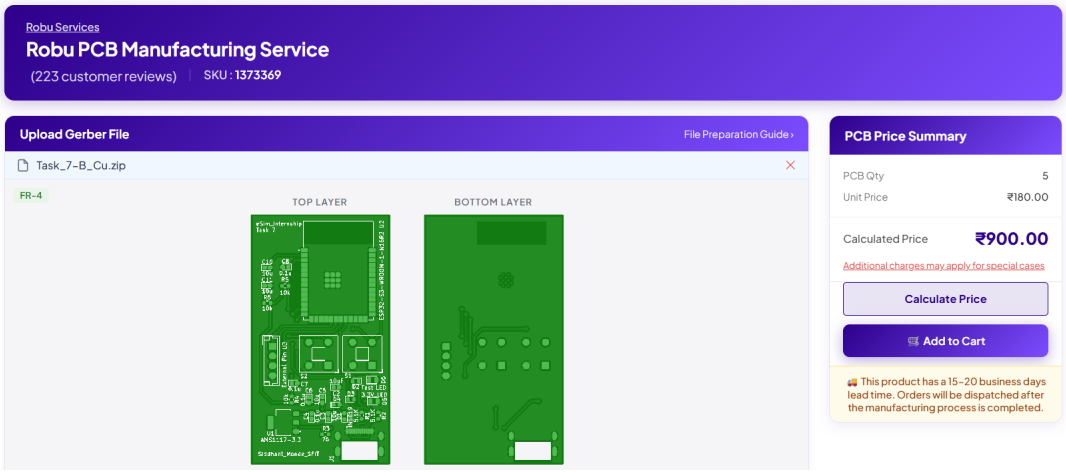


Figure 12.1: PCB Manufacturing Preview

Chapter 13

3D View

13.1 PCB 3D Front View

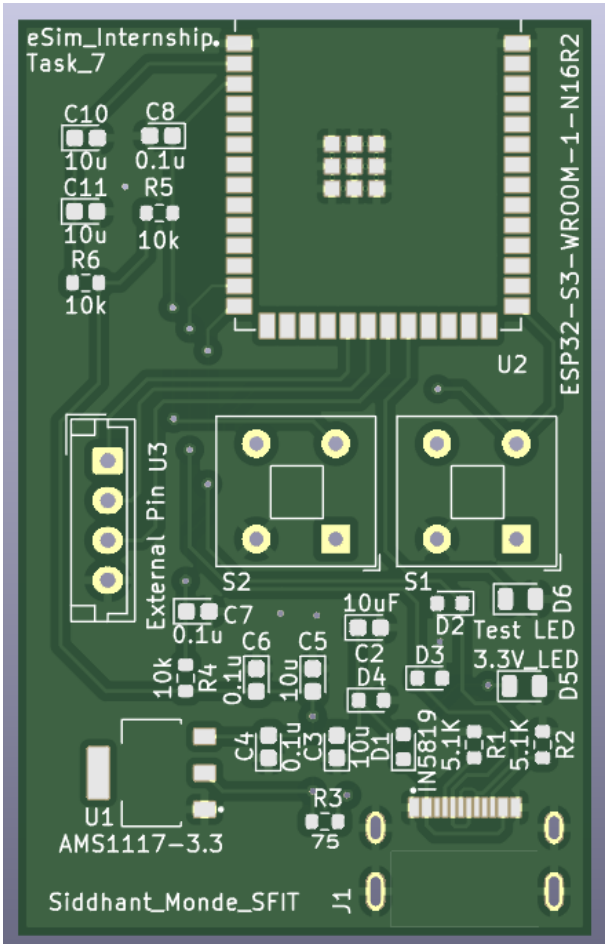


Figure 13.1: PCB 3D Front View

13.2 PCB 3D Rear View

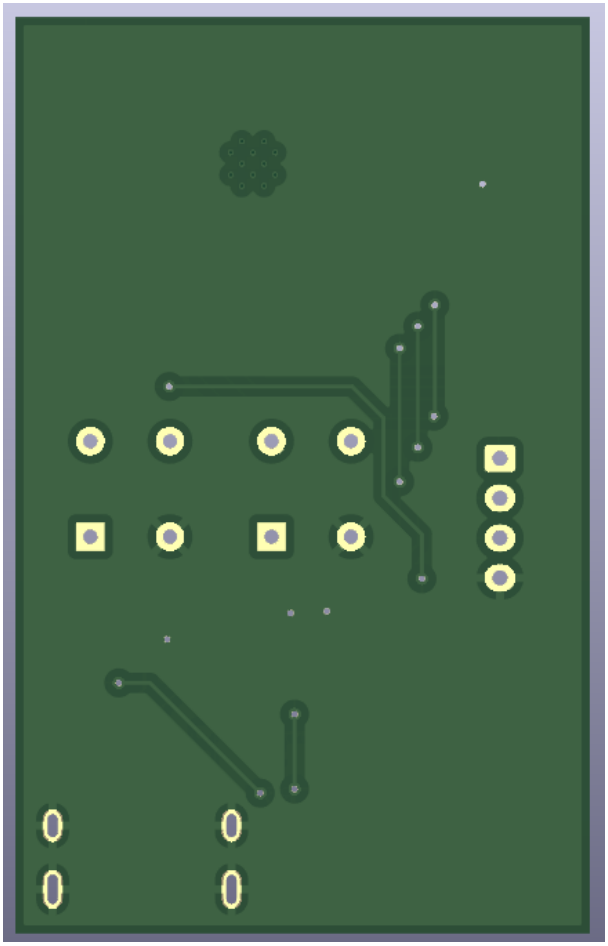


Figure 13.2: PCB 3D Rear View

Chapter 14

Conclusion

The internship/technical task provided practical exposure to the complete process involved in electronic circuit and PCB development.

As part of the work associated with the **eSim IIT Bombay software ecosystem**, a custom ESP32-S3-based SMD PCB was designed from the initial circuit concept through schematic development, component selection, footprint assignment, PCB placement, routing, verification, and manufacturing preparation.

The developed design integrates USB Type-C power and communication, 5 V to 3.3 V power regulation, ESP32-S3 control, boot and reset functionality, GPIO expansion, test LEDs, protection components, and local decoupling.

A significant part of the task involved understanding the practical reasoning behind PCB design decisions. Factors such as capacitor selection and placement, SMD package availability, pull-up resistor selection, trace width, via usage, grounding, routing, and manufacturing constraints were considered throughout the design process.

The task also provided practical experience with electronic design automation tools and the PCB development workflow, including schematic-level design, layout development, routing, ERC and DRC verification, and manufacturing preparation.

Overall, the work successfully strengthened the understanding of practical electronic hardware development and demonstrated the complete process required to convert an electronic circuit concept into a manufacturable PCB.

The experience gained through this task provides a strong foundation for future work involving embedded systems, robotics controllers, and advanced PCB design.

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